Walter Andrade

Senior Software Engineer, AI Products | TypeScript, Python, React | LLM agents, structured outputs, evals

Belo Horizonte, Minas Gerais, Brazil

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Top skills: TypeScript · AI Agents · Node.js · React · PostgreSQL

Summary

Senior software engineer with 6 years of TypeScript, React and Node.js and about a year shipping LLM features to production. At SmartHow, a compliance SaaS for heavy industry, I designed and built the workspace's AI agent from scratch, which cut writing a new document from 3 hours to 10 minutes; shipped document generation from PDFs and audio with structured outputs, repair and evals; and built a voice-agent platform on Recall.ai and ElevenLabs. My rule: the model proposes; code and people guarantee the contract.

Experience

SmartHow

Sep 2025 – Aug 2026 · 1 yr

Senior Software Engineer

Belo Horizonte, Brazil

Multi-tenant knowledge and compliance SaaS for heavy industry (ISO 9001/14001/45001).

- Top PR author on a team of four to six developers: 473 merged PRs (42.7% of all team merges) across frontend, backend, AI and infrastructure, and de facto release manager for the production product (59 of 112 release promotions).
- Designed and built the workspace's AI agent from scratch, replacing a chat that stopped after three tool rounds, dropped parallel tool calls and forgot its work between turns: a thin custom loop chosen over LangGraph and Temporal, with parallel tool calls, tool errors as evidence, a stop policy with a cost cap and cross-turn memory, in production since June 2026. The agent cut writing a new document from 3 hours to 10 minutes, and creation to publication from 5 days to 1 on average.
- Shipped document generation from PDFs and audio on one rule: the LLM writes, code guarantees the contract. Structured outputs, targeted repair, safe truncation and a pre-persist gate in an async BullMQ job; final eval 19 pass / 0 fail, 124 real fields, none over limit. Fixed a production content regression by A/B-testing old prompts recovered from git (0 → 23 of 23 steps).
- Built a voice-agent platform that interviews industrial operators: a meeting bot on Recall.ai and ElevenLabs in two weeks, then NestJS orchestration where each meeting or call is a deployment state machine with signed webhooks (Svix, HMAC) and BullMQ scheduling. Built a Twilio Media Streams bridge in a day and swapped it for ElevenLabs native telephony six days later; a port/adapter split kept the swap cheap. The voice agents handled 724 interactions: 447 phone calls and 277 video meetings, 55+ hours in total.
- Built asset ingestion and retrieval for the workspace agent: BullMQ queue, Python extraction, chunks in Postgres full-text search. Added a router that decides before the model whether a turn needs sources, a fallback that reads the original PDF when chunks fail, and source-based edit suggestions with an evidence pack and human accept/reject. Re-scoped assets per document; the isolation eval showed zero cross-document leakage.
- Co-built a Projects module where auditors run audits with an AI agent that reads only authorized scope and only proposes writes: backend commands, the full original web UI, PDF report export and the ADR, in 35 PRs over five days. Before calling it done, ran 13 real journeys with browser agents against a live LLM; they found 2 critical flaws review and tests had missed (the agent saw every nonconformity in the tenant, and closed projects accepted writes), both fixed the next day.
- Built AI checklists from scratch across three services (33 PRs, in production since December 2025): a LangGraph pipeline that turns a procedure into one section per step with structured-output questions, and photo reading of paper checklists by a vision model, with per-question confidence and a review screen before saving.
- Made the agent legible: an additive WebSocket frame contract (ADR) so backend and frontend deploy in any order, an error classifier with category, retry and correlation id, and live agent activity instead of a minute of three dots. Replaced a random prompt-injection delimiter that broke prefix caching with a per-session HMAC: ~76% cache reads across 25 traced agent runs.
- Ran coding agents like a team: PRD, vertical issues with acceptance criteria, one subagent per issue in its own git worktree, then review and merge of every diff. Added systematic distrust: parallel multi-lens review that caught a foreign-key ordering bug Greptile and tests missed, and 7 investigator agents against 7 skeptics on 7 pilot bugs, with no diagnosis overturned and the fix shipped the same day.

Instaq – Instituto Taqtile

Oct 2023 – Sep 2025 · 2 yrs

Senior Full Stack Web Developer | Mentor | Product Owner

Brazil

Senior developer in a group of developers in training serving real clients: product owner and main client contact on several products at once, and mentor to junior developers. **Modulart:** sole author of the Payload CMS 3 backend (Next.js 15, PostgreSQL). **NaPorta:** Next.js landing page and Refine admin; the project won a national competition. **Payssego:** led a Node.js/TypeScript salary-advance platform; 1,100+ users live on day one. Also **Vivenda** (Astro, AWS EC2) and **Pence** (React, Node.js, Neo4j).

Taqtile Brasil

Sep 2020 – May 2023 · 2 yrs 9 mos

Software Engineer

São Paulo, Brazil

Grupo BIG: frontend of the mobile retail app for Sam's Club, Walmart, Bompreço and Hiper Bompreço (Sam's Club Brasil: 1M+ Google Play downloads). **Heringer:** web side of the company's website. Also the **ioP Pet** React Native app (published to both stores), an internal Redmine replacement with BigQuery and Looker Studio dashboards, **PoliAngels** and the **Gentelab** e-learning frontend (React, Relay, Hasura).

Skills

AI & agents: OpenAI, Azure OpenAI and Anthropic APIs · Agent loops and parallel tool calling · Structured outputs · Evals and browser test batteries · Retrieval and PDF ingestion (Postgres full-text search) · LangGraph · LangSmith tracing · MCP servers and harnesses · Prompt caching and model routing · Voice agents (ElevenLabs, Recall.ai, Whisper) · Fine-tuning (PyTorch, transformers), ONNX serving, Hugging Face · Claude Code, coding agents in worktrees, multi-agent review

Engineering: JavaScript, TypeScript, Node.js, React, React Native, Next.js, React Query, Zustand, Zod, GraphQL, REST APIs, PostgreSQL, SQL, Neo4j, CI/CD, Vitest, Playwright, Docker, Git, Figma, Front-End & Back-End Development, State Management, OOP.

Education

Universidade Federal da Bahia

2014 – 2019

Bachelor's degree, Philosophy of Mathematics | Logic

Labenu

2020

Web Full Stack Development, Information Technology

Certifications

Google

Sep 2026

AI Fundamentals, AI for Brainstorming and Planning, AI for Research and Insights, AI for Writing and Communicating, AI for Content Creation, AI for Data Analysis, AI for App Building, AI for App Deployment

Languages

Portuguese (Native) · English (Native / Bilingual) · Spanish (Elementary)